EXPERIMENT 6: Install Windows in VirtualBox and Find the Largest Element in an Array Using C

1. Aim

To install Windows in Oracle VirtualBox and execute a C program to find the largest element in a given array.

2. Procedure

1. Install Oracle VirtualBox on the host computer.
2. Download a Windows ISO image from the official Microsoft website.
3. Open VirtualBox and click New to create a virtual machine.
4. Enter the virtual machine name and select Microsoft Windows as the operating system type.
5. Allocate RAM and create a virtual hard disk with sufficient storage.
6. Attach the Windows ISO image and start the virtual machine.
7. Follow the Windows setup instructions and complete the operating system installation.
8. Install a C compiler such as MinGW-w64 or an IDE bundled with a compiler.
9. Open the compiler terminal and create a file named largest.c.
10. Enter the C program to read array elements and find the maximum value.
11. Compile the program using GCC and execute the generated program.
12. Enter the array size and elements, then verify the largest value.

3. Program

Program – largest.c:

```
#include <stdio.h>

int main()
{
    int a[100], n, i, max;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    if (n < 1 || n > 100) {
        printf("Invalid array size");
        return 1;
    }

    printf("Enter array elements:\n");
    for (i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    max = a[0];

    for (i = 1; i < n; i++) {
        if (a[i] > max) {
            max = a[i];
        }
    }

    printf("Largest element = %d\n", max);
    return 0;
}
```

Compile and execute in the compiler environment:

```
gcc largest.c -o largest.exe
```

```
largest.exe
```

4. Output of Program

```
Enter number of elements: 5
Enter array elements:
10
25
45
12
30
Largest element = 45
```

5. Result

Thus, Windows was installed in VirtualBox, and the C program to find the largest element in an array was compiled and executed successfully.

EXPERIMENT 6: Install Windows in VirtualBox and Find the Largest Element in an Array Using C

1. Aim

To create a Windows virtual machine using Oracle VirtualBox, install the Windows operating system and execute a C program to identify the largest element in a given array.

2. Procedure

1. Download and install Oracle VirtualBox on the host computer. 2. Download a compatible Windows ISO image from the official Microsoft website. 3. Open VirtualBox and click the New option to create a virtual machine. 4. Enter the virtual machine name and select Microsoft Windows as the operating system type. 5. Allocate sufficient RAM and processor resources to the virtual machine. 6. Create a virtual hard disk with adequate storage space for the Windows installation. 7. Attach the downloaded Windows ISO image to the virtual optical drive. 8. Start the virtual machine and wait for the Windows installation screen to appear. 9. Select the language, time format and keyboard layout. 10. Follow the Windows installation wizard and complete the operating system setup. 11. Create a user account and configure the initial Windows settings. 12. Restart the virtual machine and verify that Windows starts successfully. 13. Install a C compiler such as MinGW-w64 or an IDE that includes a GCC compiler. 14. Open the compiler terminal or command prompt and create a file named largest.c. 15. Enter the C program to read the array size and its elements from the user. 16. Initialize the maximum variable with the first array element. 17. Compare each remaining array element with the current maximum and update it whenever a larger value is found. 18. Compile the program using GCC and execute the generated executable file. 19. Enter the array size and sample values when prompted. 20. Observe the output and verify that the program displays the largest element correctly.

4. Output of Program

Sample Input: Enter number of elements: 5 Enter array elements: 10 25 45 12 30 Expected Output: Largest element = 45 Explanation: The program reads all elements of the array and compares them one by one. The maximum variable stores the largest value encountered during the comparison process. After all elements have been checked, the largest value is displayed.

5. Result

Thus, Windows was successfully installed in Oracle VirtualBox, and a C compiler was configured in the virtual environment. The C program to find the largest element in an array was compiled and executed successfully, producing the correct result.